Applications of similar triangles

Core Concepts: Applications of similar triangles

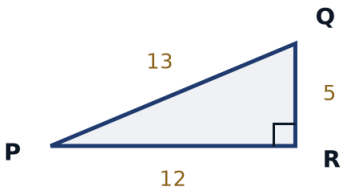
1 In $\triangle ABC$ studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 96°

b. 106°

c. 88°

d. 141°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm

c. 24 cm

d. 32 cm

3 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 15

c. 13

d. 17

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 89°

b. 99°

c. 81°

d. 141°

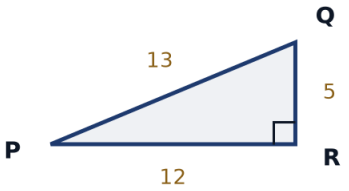
5 A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length?

a. 21 cm

b. 24 cm

c. 32 cm

d. 72 cm



6 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 13

b. 19

c. 17

d. 21

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 82°

b. 92°

c. 74°

d. 141°

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8 A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm

c. 40 cm

d. 128 cm

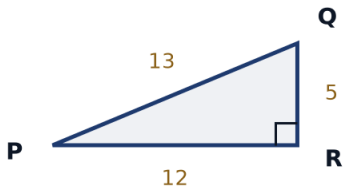
9 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$?

a. 17

b. 23

c. 21

d. 25



Exam-Style Problem Solving & Modeling

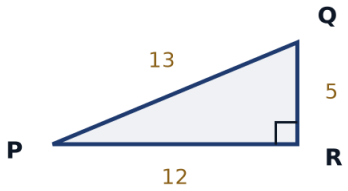
10 In $\triangle ABC$ studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 75°

b. 85°

c. 67°

d. 141°



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11 A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm

c. 48 cm

d. 200 cm

12 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 4$?

a. 21

b. 27

c. 25

d. 29

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